

An illustration of an Atlas rocket launching from the bottom left, with a large plume of orange and yellow fire and smoke. Several pieces of paper with handwritten notes are floating around the rocket. In the upper right, the Gemini-Agena Target Vehicle (GATV) is shown in space, suspended by a thin line. The background is a deep purple space with white stars and a large, curved, light-colored horizon line.

Mary worked on the Gemini-Agena Target Vehicle (GATV). Its most important job was helping spacecraft to dock safely with others in space.

To the stars

Mary was brought into a group of engineering experts at Lockheed, known as Skunk Works. At first, the wonder-team was tasked with designing missiles and defense systems. It was groundbreaking work, and Mary was also making history herself, as both the first Native American aerospace engineer and the only woman in the group of 40 engineers.

The US then set its sights on space, and Mary used mathematics to help figure out how to send a satellite into space for the first time. She did calculations to predict how the spacecraft would behave when moving through the Earth's atmosphere, and how to make sure it stayed on the right path. Later, Mary helped to plan for missions to Mars and Venus. Even now, her work is still classified (top secret)!

Lifting off

The GATV was carried into space by an Atlas rocket. Once in space, NASA used it to plan and test maneuvers such as changing orbit. This was important preparation for the Apollo program, which took astronauts to the moon.

Mary's life work was honored by a special \$1 coin.